

1. Introduction to Blockchain

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C O N T E N T S

1. "Crypto"

What is Blockchain?

Block and Chain

2. Cryptographic Mechanisms

Mining

The Longest-chain Rule

Identification

Transaction Verification

Proof of Stake vs Proof of Work

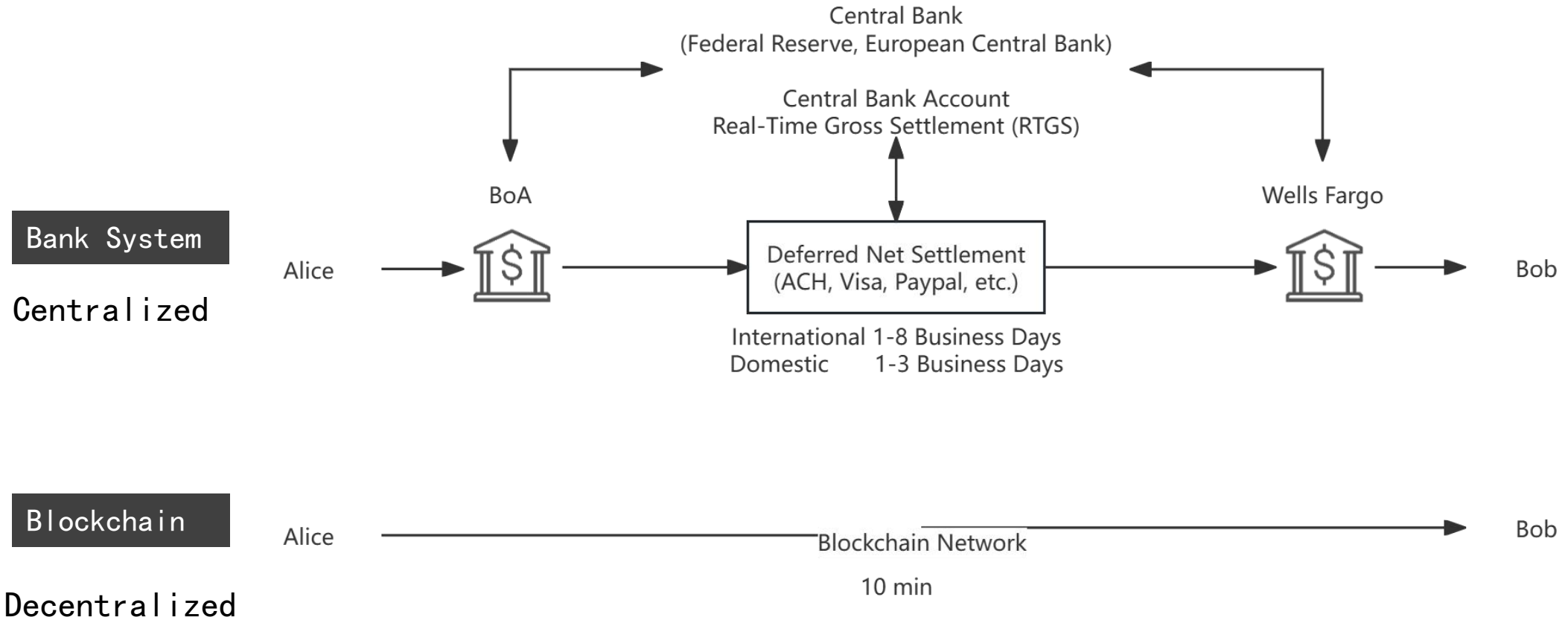
3. "Currency"

What is a Currency?

1. “Crypto”

First Impression

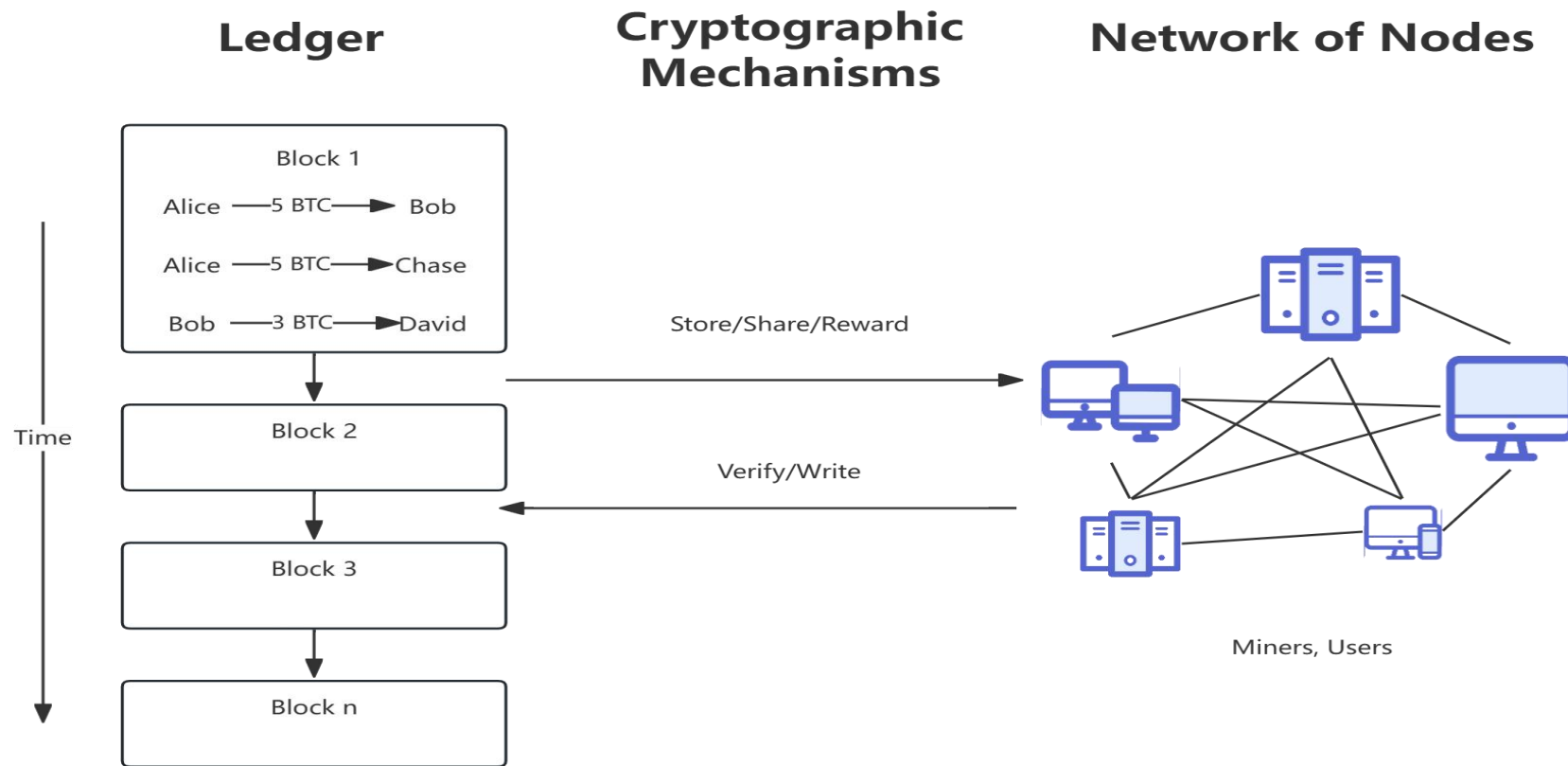
Payment Renovation: Traditional Bank vs. Blockchain



- FedNow (Federal Reserve) and RTP (Clearing House) are non-blockchain real-time payment services (only available now to U.S. banks).
- Blockchain is much more efficient for international transactions.
- There's no institutions, regulators needed for on-chain payment.

1. “Crypto” : What is Blockchain?

Blockchain is a digital ledger which is maintained across computers linked in a peer-to-peer network.



1. “Crypto” : Block and Chain

Ledger (Blocks)

1 Block $\approx$ 4000 Transactions (1 Mb)

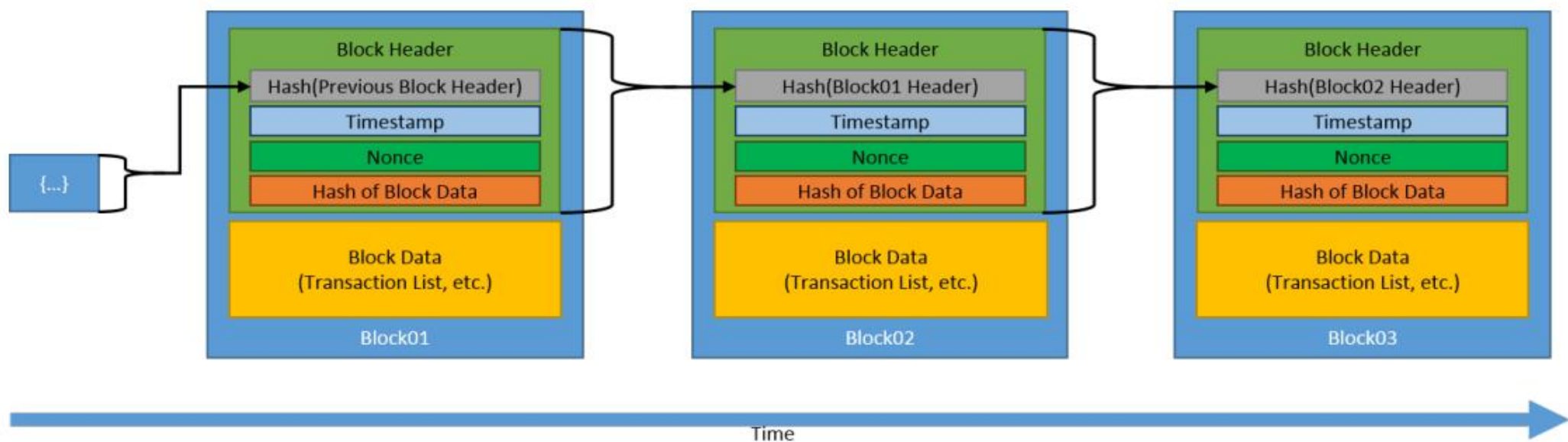
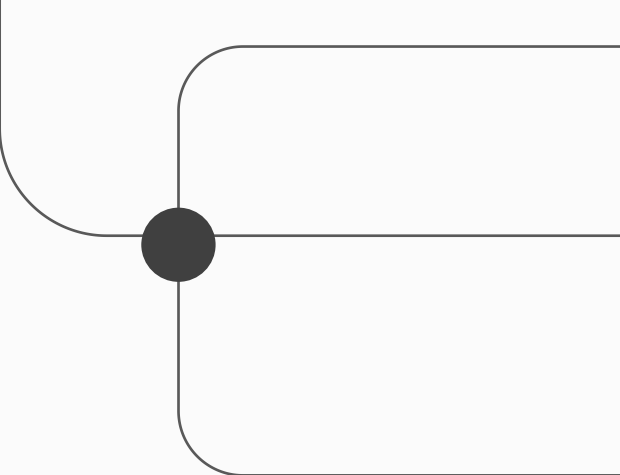
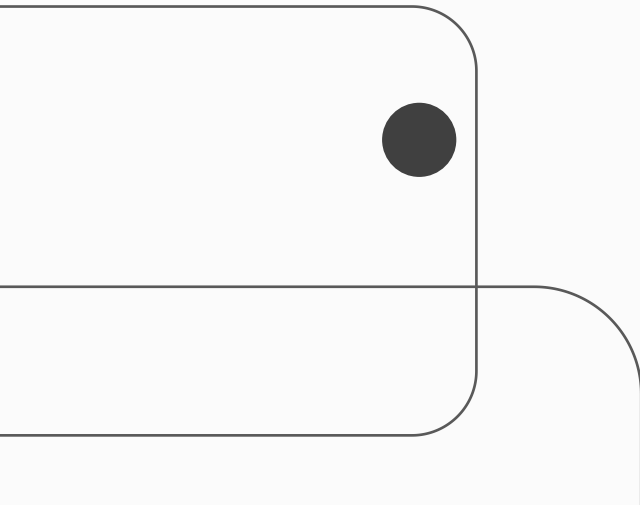


Figure.Generic Chain of Blocks (Yaga et al. 2019)



2. Cryptographic Mechanisms



2. “Cryptographic Mechanisms” : The Puzzle

Consensus Algorithm

Bitcoin uses the Proof-of-Work (PoW) consensus algorithm. The first one to solve the puzzle gets to add the next block to the chain and is rewarded with bitcoins.

The Puzzle:

$\text{SHA256}(\text{“blockchain”} + \text{Nonce/Random Number}) = \text{Hash Digest starting with “000000”}$

```
SHA256("blockchain0") = 0xbd4824d8ee63fc82392a6441444166d22ed84eaa6dab11d4923075975acab938 (not solved)
SHA256("blockchain1") = 0xdb0b9c1cb5e9c680dfff7482f1a8efad0e786f41b6b89a758fb26d9e223e0a10 (not solved)
...
SHA256("blockchain10730895") = 0x000000a1415e0bec568f6f605fcc83d18cac7a4e6c219a957c10c6879d67587 (solved)
```

SHA-256: is a cryptographic hash function whose core characteristic is its one-way nature—making it computationally infeasible to retrieve the original input from the resulting hash. It took 934,224,175 guesses to solve the example puzzle.

- The current BTC difficulty is 127,620,086,886,390 (127T) at block 909,044.
- Every 2016 blocks, the BTC network will change its difficulty (around every 2 weeks).

Details in Article for Bitcoin Difficulty: <https://en.bitcoin.it/wiki/Difficulty>

2. “Cryptographic Mechanisms” : Mining

Rewarding Rules:

1. A new block will be created every 10 mins.
2. The first puzzle solver is the only one to wrap the block.
3. 50 BTC will be rewarded to the puzzle solver.(50% reduction every 4 years, around 3 BTCs for now)

Total Supply of Bitcoins

$$50 \times 6 \times 24 \times 365 \times 4 \times (\sum_{i=0}^{\infty} 0.5^i) \approx 21 \text{ million BTC}$$

Block < 909048 >			
Hash	000000...0f07870	Fee span	2.00 - 262 sat/vB
Timestamp	2025-08-07 13:02:08 (13 minutes ago)	Median fee	~3 sat/vB \$0.49
Size	1.63 MB	Total fees	0.038 BTC \$4,378
Weight	3.99 MWU	Subsidy + fees	3.163 BTC \$367,772
Health ⓘ	100%	Miner	⚡ Luxor

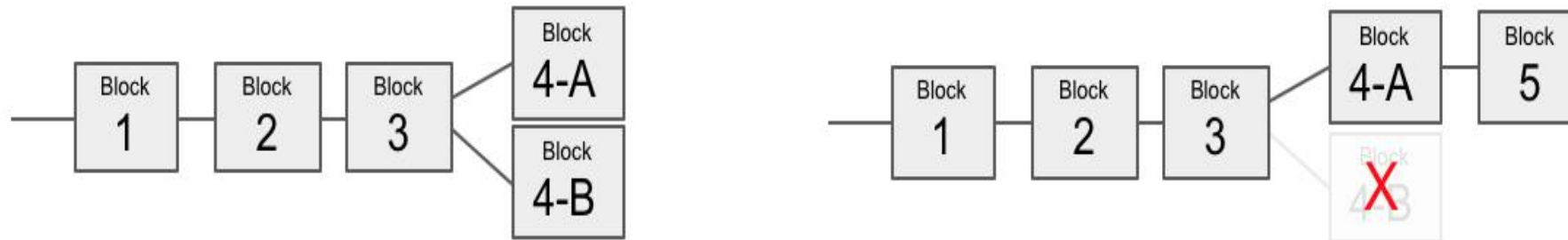
source:mempool.space

2. “Cryptographic Mechanisms” : The Longest-chain Rule

How to avoid attacks such as double-spending?

The Longest-chain rule:

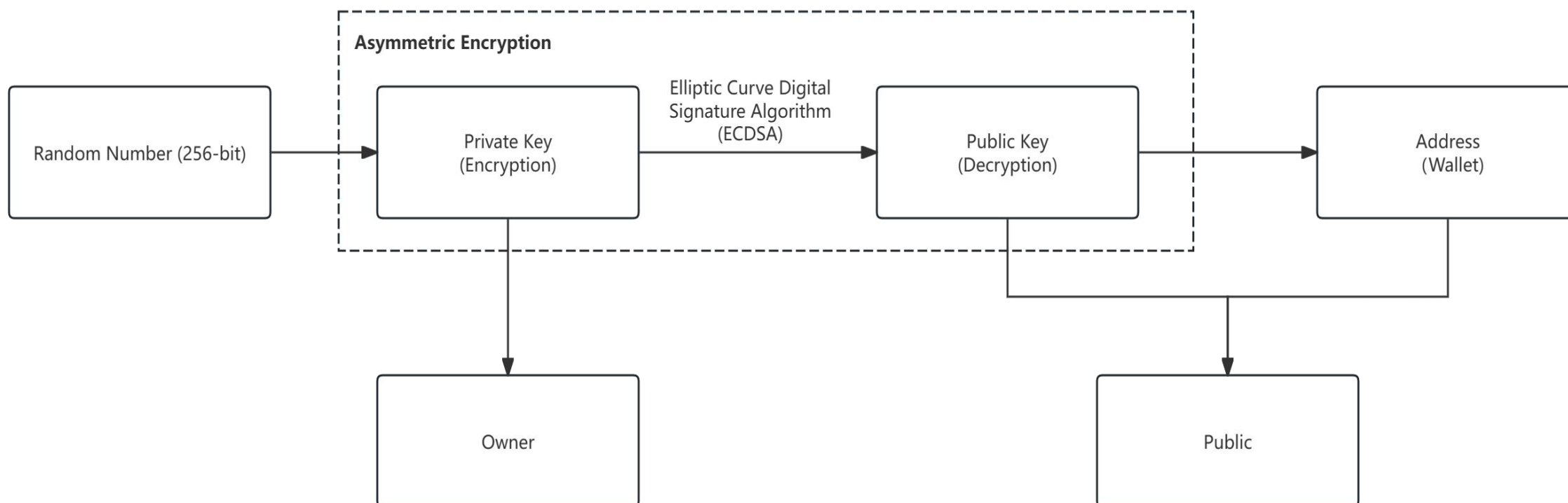
The longest chain is the version of the blockchain that nodes recognize as valid, enabling consensus on the transaction history.



Resolving conflicts - The longest chain wins

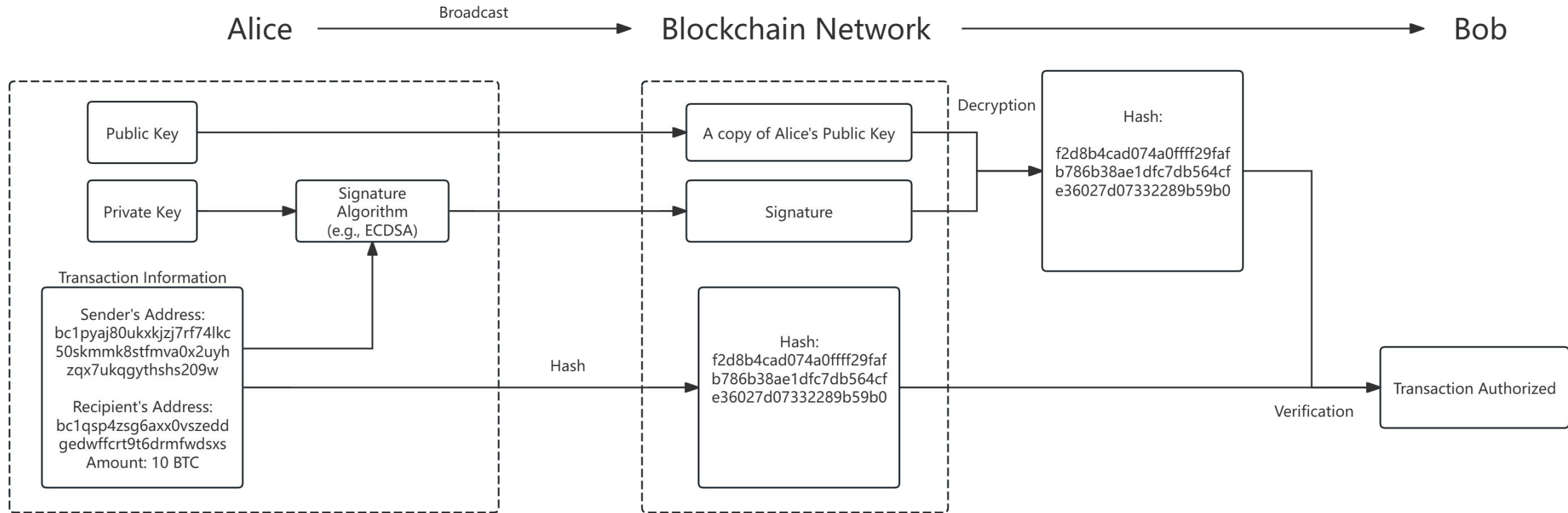
2. “Cryptographic Mechanisms” : Identification

How a crypto wallet is created and encrypted?



2. “Cryptographic Mechanisms” : Transaction Verification

How transactions are verified on blockchain?



2. “Cryptographic Mechanisms” : Mining (solo mining)

- It is almost impossible for a single mining machine to solve the puzzle.
- Bitcoin Miner S21e XP Hyd (\$13,201) calculates 430 trillion hash per second (Th/s), with the current Bitcoin hashrate is 1.076 ZH/s (1 ZH=1 million TH).
- A solo miner must attempt approximately 279,069 blocks to successfully mine one block. (5.3 years).
- break even in 3 years (theoretically)

Risk for mining:

- increase of electricity price
- increase of network difficulty
- decrease of BTC price

BITMAIN

ANTMINER

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Cryptocurrency

Preset Coin Price

Preset Difficulty

BTC

118076.82

\$

120435235580344.9

ANTMINER Mining Server

Power Consumption

Hashrate

Miner Cost

Switch

Quantity of Miners

S21e XP Hyd.

5500

W

430.00

T

13200

\$/Unit

1

Electricity Fee

Other Costs

Mining Period

8.15

/kWh

0

\$(Day/Miner)

2025-08-14

-

2026-08-14

Calculate

Expected Return

Days To Payback

1089.55

Day

Mining Days

365.00

Day

Actual Return On Investment

33.50

%

Expected Profit

Total Price of Miners

\$13200

Shutdown Price

\$55466.67

Period	Mining Earnings	Electricity Consumption	Electricity Costs	Electricity Ratio	Other Costs	Daily Net Profit
Current Daily Profit	0.00019350 BTC \$ 22.85	134.16 kWh	\$ 10.73	48.98%	\$ 0.00	\$ 12.12
Mining Period Profits	0.07062750 BTC \$ 8339.47	48968.40 kWh	\$ 3917.47	48.98%	\$ 0.00	\$ 4422.00

2. “Cryptographic Mechanisms” : Mining (mining pool)

- A mining pool is the pooling of resources by miners, who share their processing power over a network, to split the reward by shares.
- The break even BTC price now for Antpool is around 118,000 USD.



English ▼

Cloud Mining Product

ANTPOOL does not make representations regarding estimated earnings

Static Output Ratio 100.06%

BTC Price Estimate

Reference Coin Price \$128,396



Current BTC Price: \$118,252.93

Earnings Estimate

Input costs 544.32 USD

Estimated daily income ⓘ 0.00001178 BTC

Total actual earnings 0.00001178 BTC * 360 days

0.00424214 BTC

= 544.67 USD

Net profit 0.35 USD

Static Output Ratio 100.06%

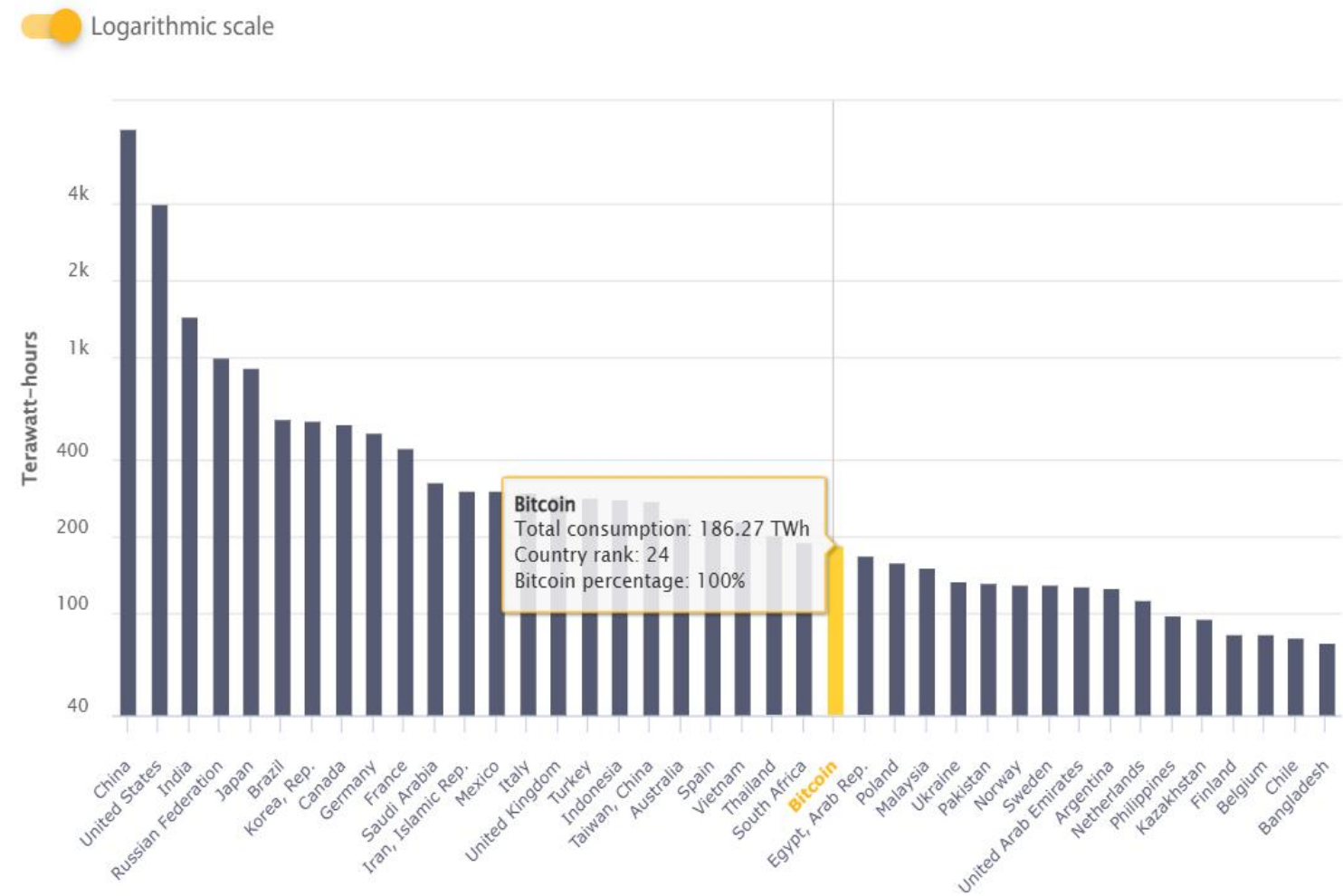
Package Details

Miner: S19 XP	Mining Coins: BTC	Mining days: Option for 150 days/240 days/270 days/360 days	Hashrate fee: Adjusted based on mining duration, longer periods result in lower hashrate fees.
Electricity cost: \$0.0413/T/Day	Validation time: Effective upon payment received.	Minimum payment: 0.005BTC	Earnings Mode: FPPS

2. “Cryptographic Mechanisms” : Energy Cost

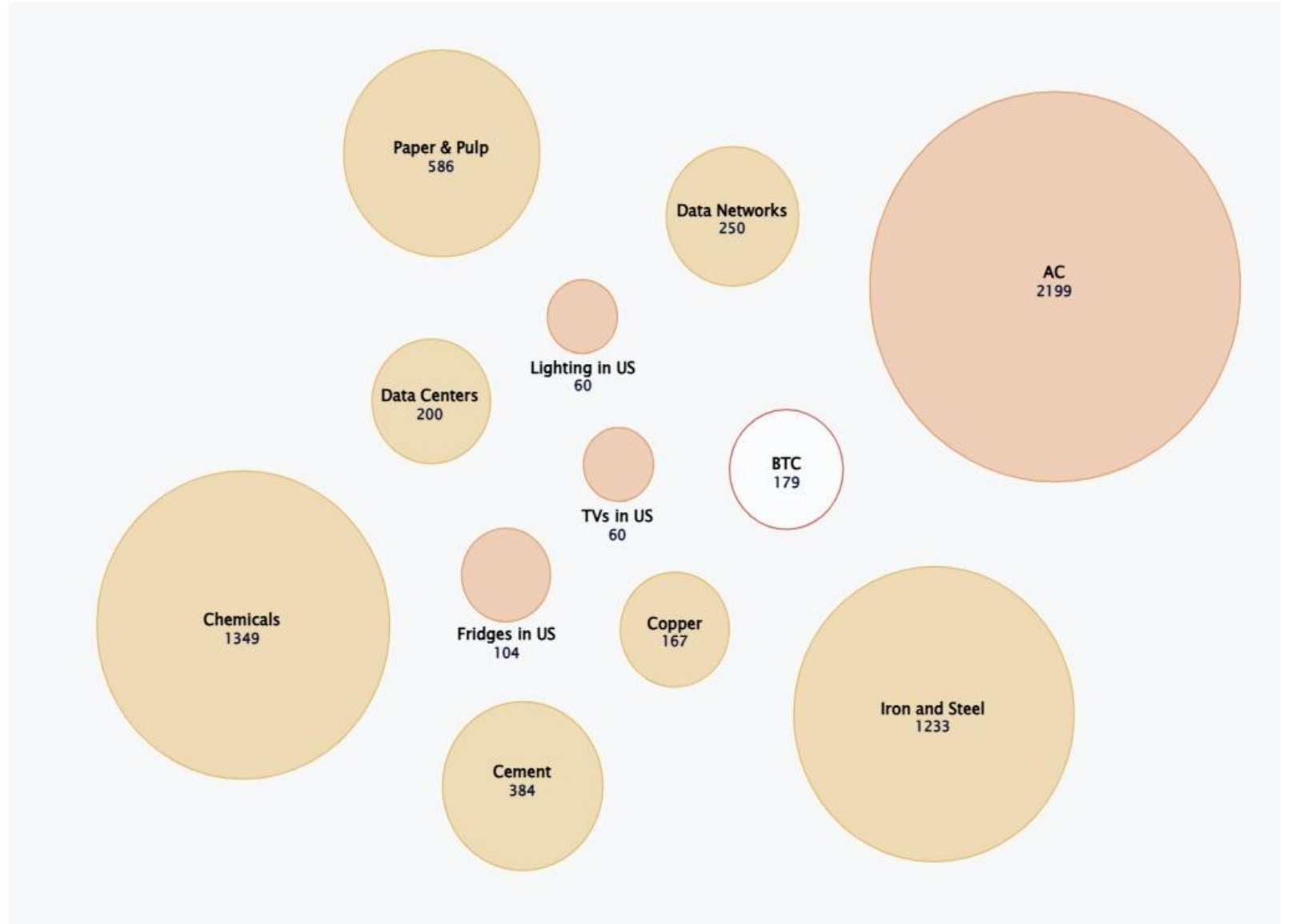
- Bitcoin is a big waste of energy!
- Bitcoin Network costs 186.27 TWh ranked 24 as a country.
- There were 1312500 BTC mined in 2024, taking the price of \$110000, the total value of BTC mined is around \$144 billion.
- The GDP of Slovakia in 2024 is \$140 billion, ranked 61 across more than 200 countries (IMF).

Country Ranking Chart



2. “Cryptographic Mechanisms” : Energy Cost

- The electricity consumed by BTC network can light up 3 United States.
- The Bitcoin network consumes 186 TWh of electricity annually, compared to 131 TWh per year for gold mining.
- BTC energy consumption is nearly equivalent to that of all data centers worldwide.



source:ccaf. io

2. “Cryptographic Mechanisms” : Valuation of BTC vs Market Price

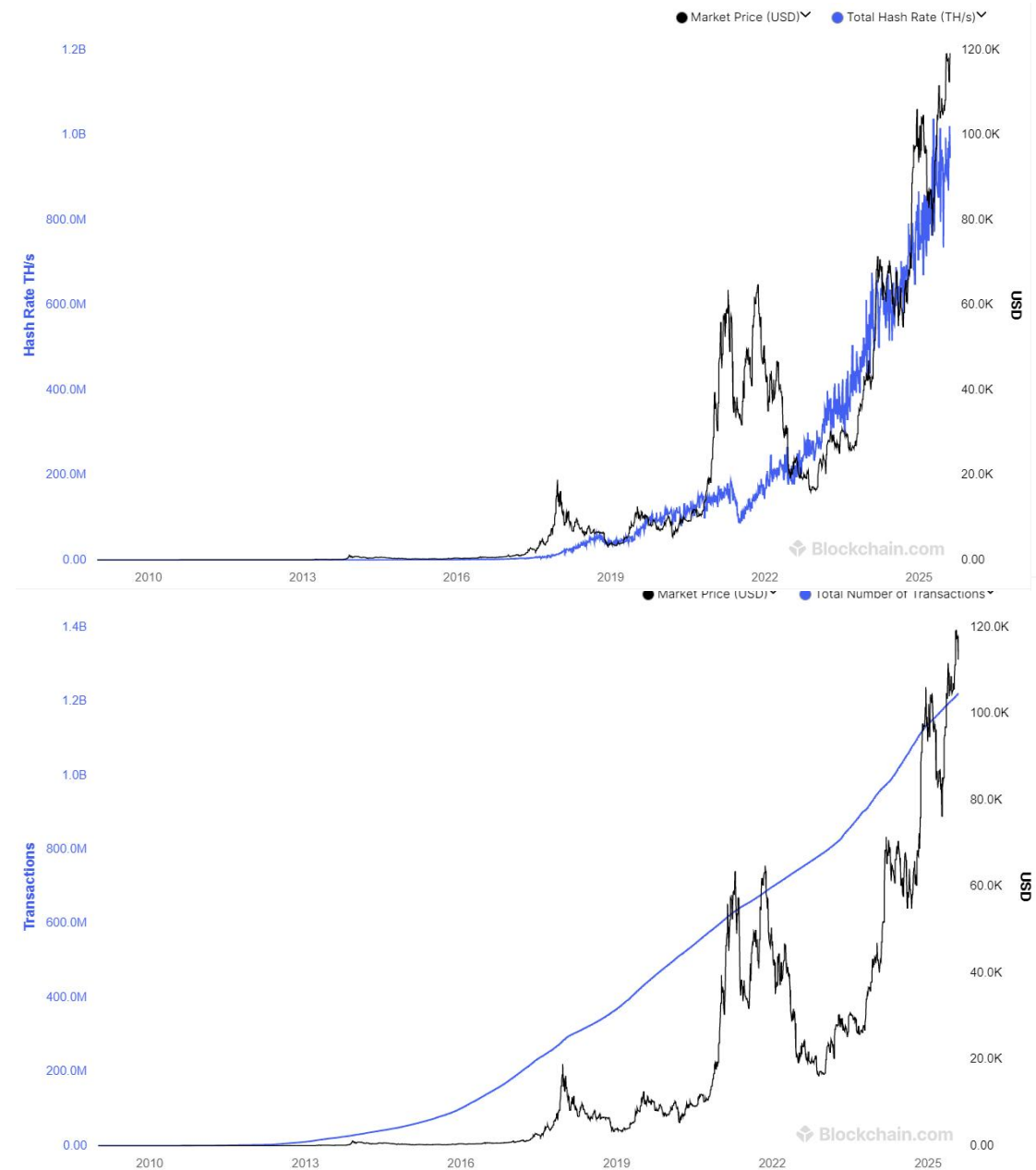
- Intrinsic Value

The Total Hash Rate reflects an increase in mining power, which raises the cost of minting new BTCs.

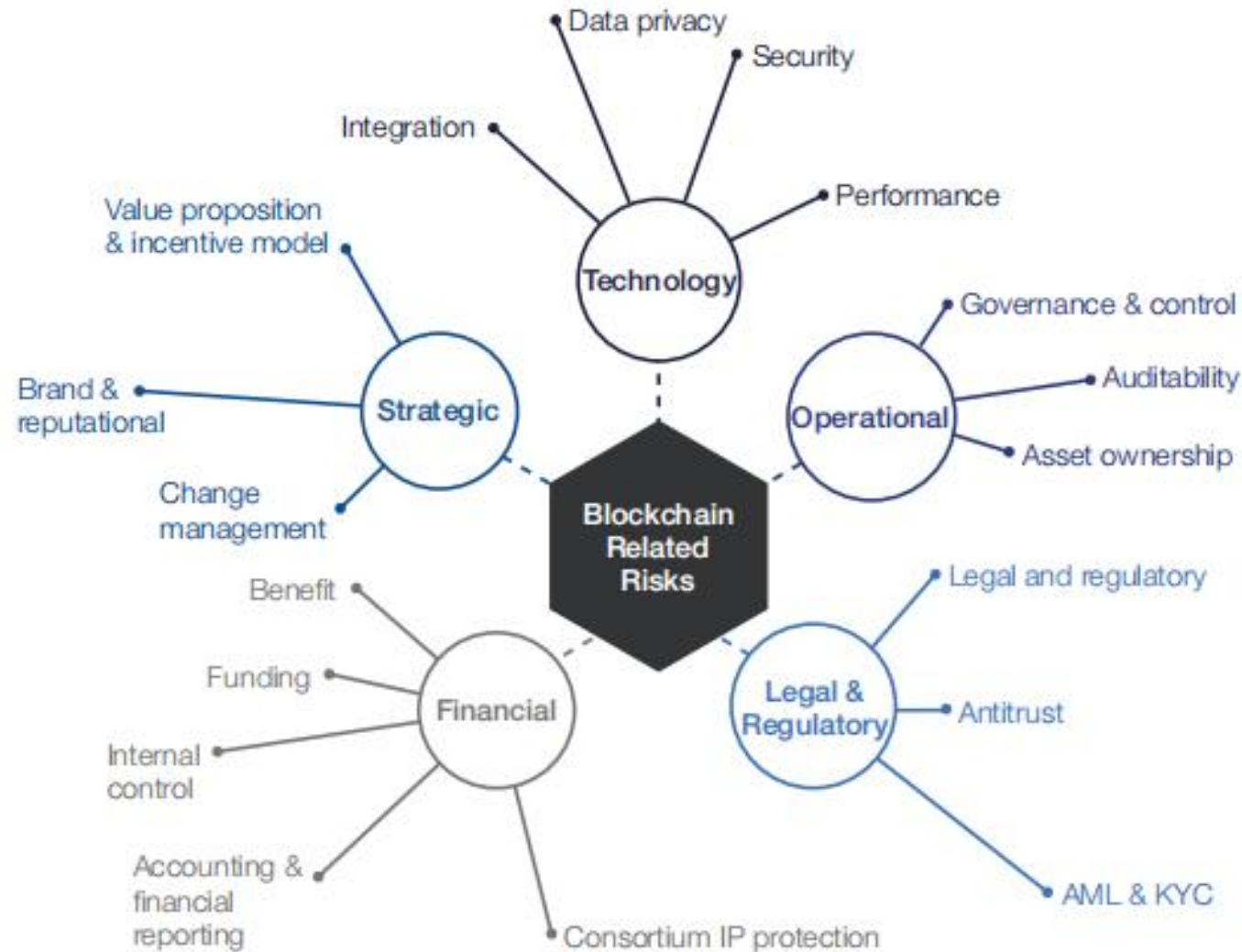
Mining power is determined by both the number of miners and the computing power of their equipment.

- Circulating Value

The total number of transactions reflects the demand for BTC.



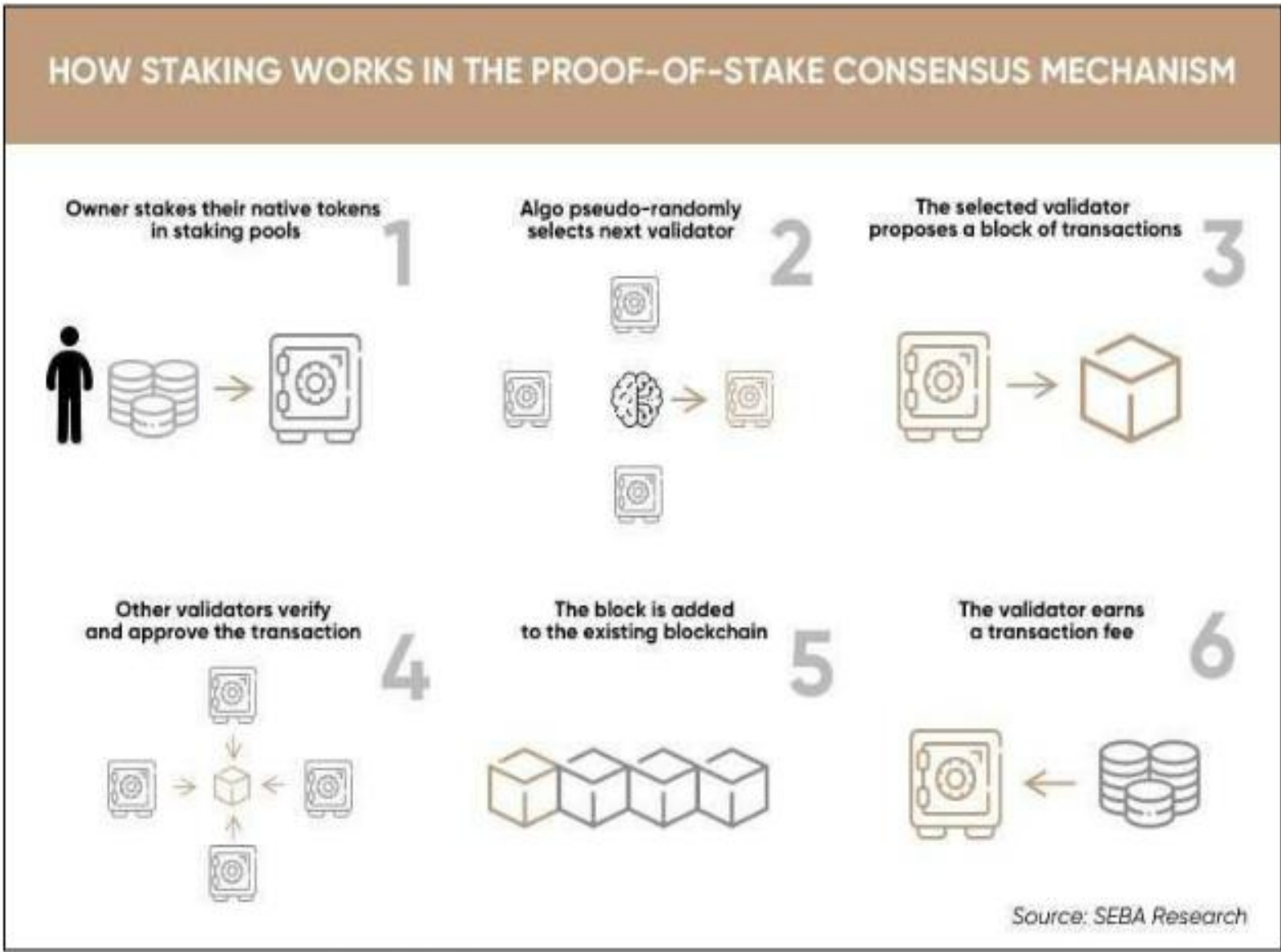
2. “Cryptographic Mechanisms” : Risk Factors



source:

<https://widgets.weforum.org/blockchain-toolkit/risk-factors/index.html>

2. “Cryptographic Mechanisms” : Proof of Stake



G2.com

Proof of Work vs. Proof of Stake		
Differences	Proof of Work	Proof of Stake
Validation method	Requires complex computational work, known as cryptocurrency mining.	The amount of cryptocurrency put at stake controls verification.
Consensus algorithm	The first to solve the cryptographic puzzles becomes the validator.	An algorithm picks a validator based on the value of their collateral.
Rewards	The first to mine the block gets a reward.	The validator earns network fee as reward.
Efficiency	PoW is less efficient due to high computational power and energy consumption.	More efficient as it doesn't require any mining equipment.
Centralization	Comes with a risk of miners with powerful equipment dominating the network.	Early adoption and wealth concentration could lead to centralized networks.
Equipment	Application-specific integrated circuits (ASICs) and graphics processing units (GPUs)	Any standard server-grade device works.
Cryptocurrency examples	Bitcoin (BTC) and Litecoin (LTC)	Ethereum 2.0 (ETH2) and Binance Coin (BNB)

3. “Currency”

3 “Currency” : Is Bitcoin a ideal currency?

Definition:

Currency is any item or verifiable record that is generally accepted as payment for goods and services and repayment of debts, such as taxes, in a particular country or socio-economic context.

Fuctions of currency:

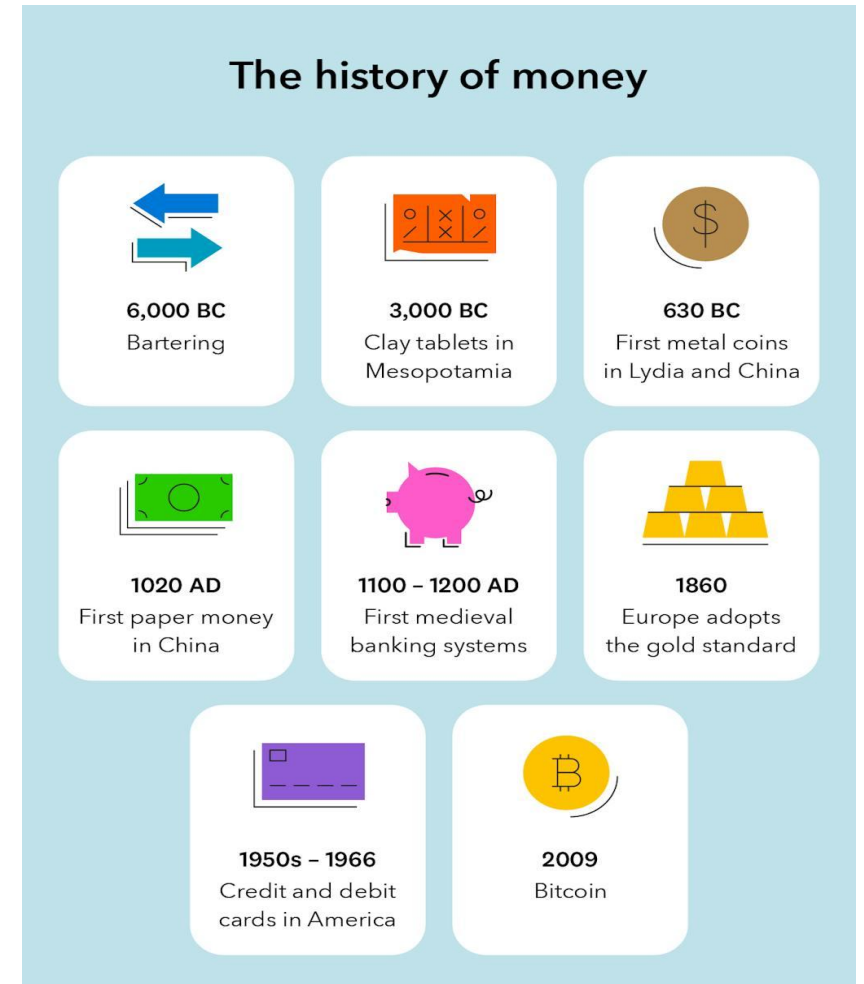
- medium of exchange
- store of value

Bitcoin' s value is too volatile to function as a currency.

- unit of account

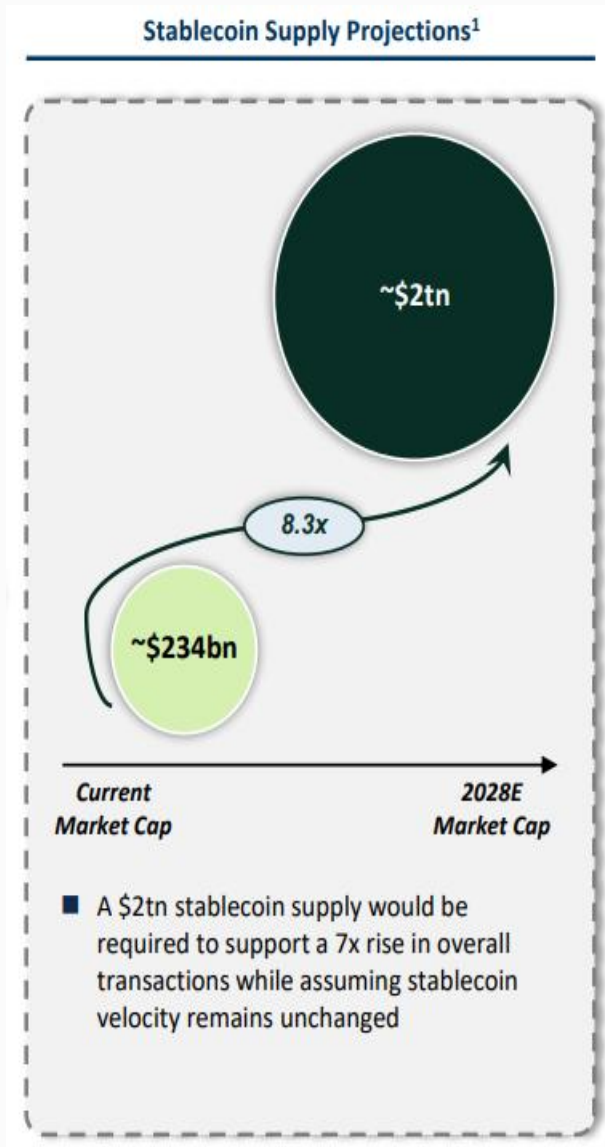
Types of Currency:

- Commodity Currency (Intrinsic Value > Circulation Value):
Shell, Gold, Grain, Cigaratte
- FIAT Currency (Intrinsic Value<< Circulation Value):
\$20 Paper Note
- Crypto Currency (Intrinsic Value<< Circulation Value):
Bitcoin? USDT? USDC?



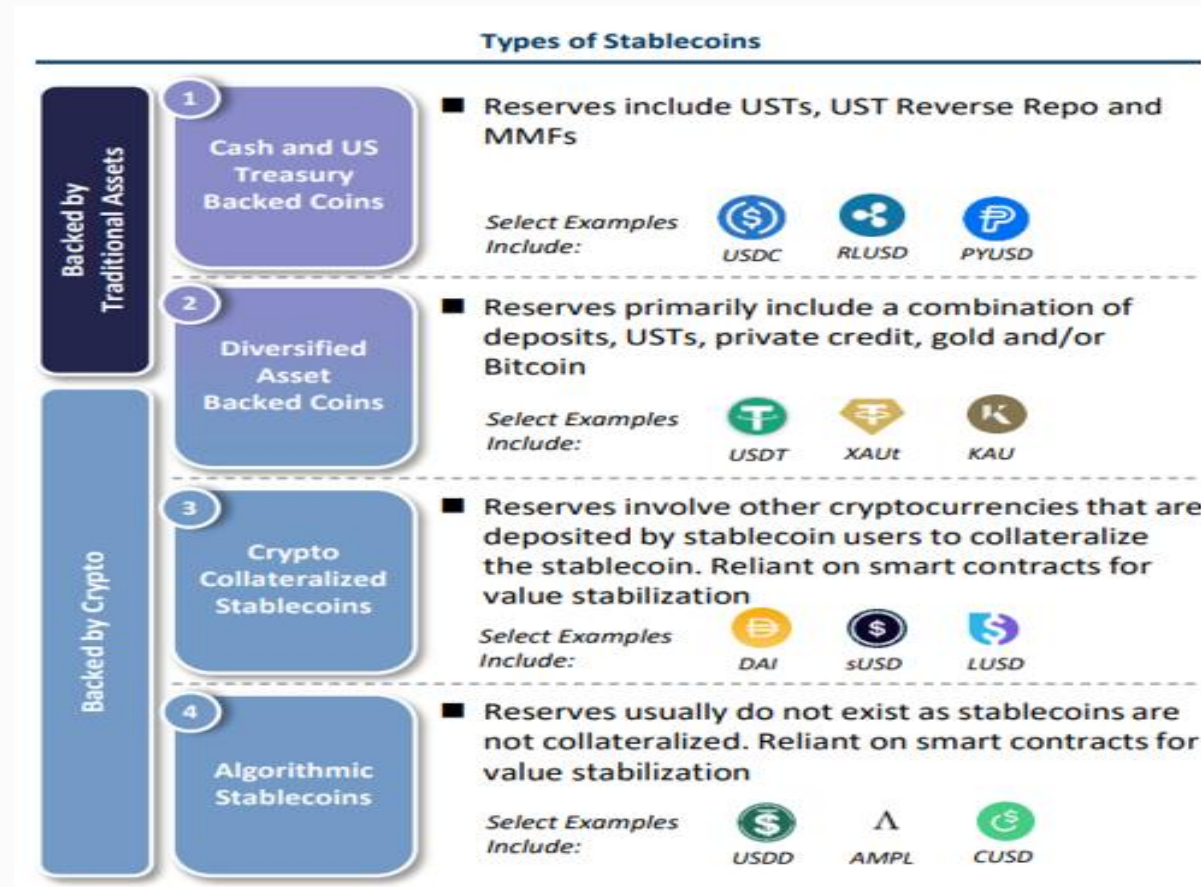
source: creditkarma.com

3 “Currency” : What is a stable coin?



Definition:

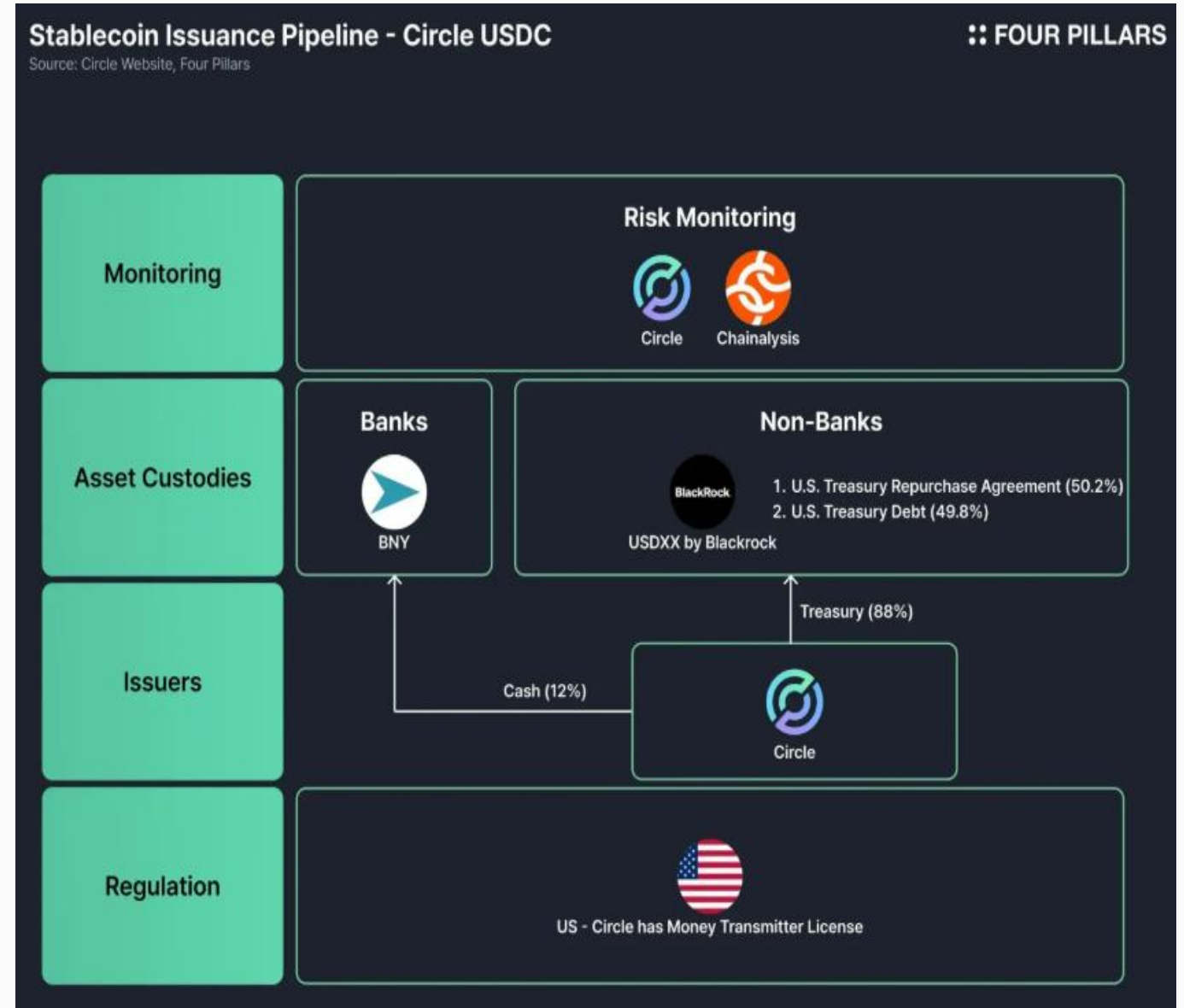
A stablecoin is a type of cryptocurrency that aims to maintain a stable value relative to a specified asset, a basket of assets.



source:treasury.gov

3 “Currency” : How stablecoins are issued?

- USDC is backed by the equivalent value of US dollar denominated assets held as reserves for the benefit of USDC holders.
- Cash is held at regulated financial institutions.
- The portfolio of the Circle Reserve Fund, which can contain short-dated US Treasuries, overnight US Treasury repurchase agreements, and cash, is custodied at The Bank of New York Mellon and is managed by BlackRock.



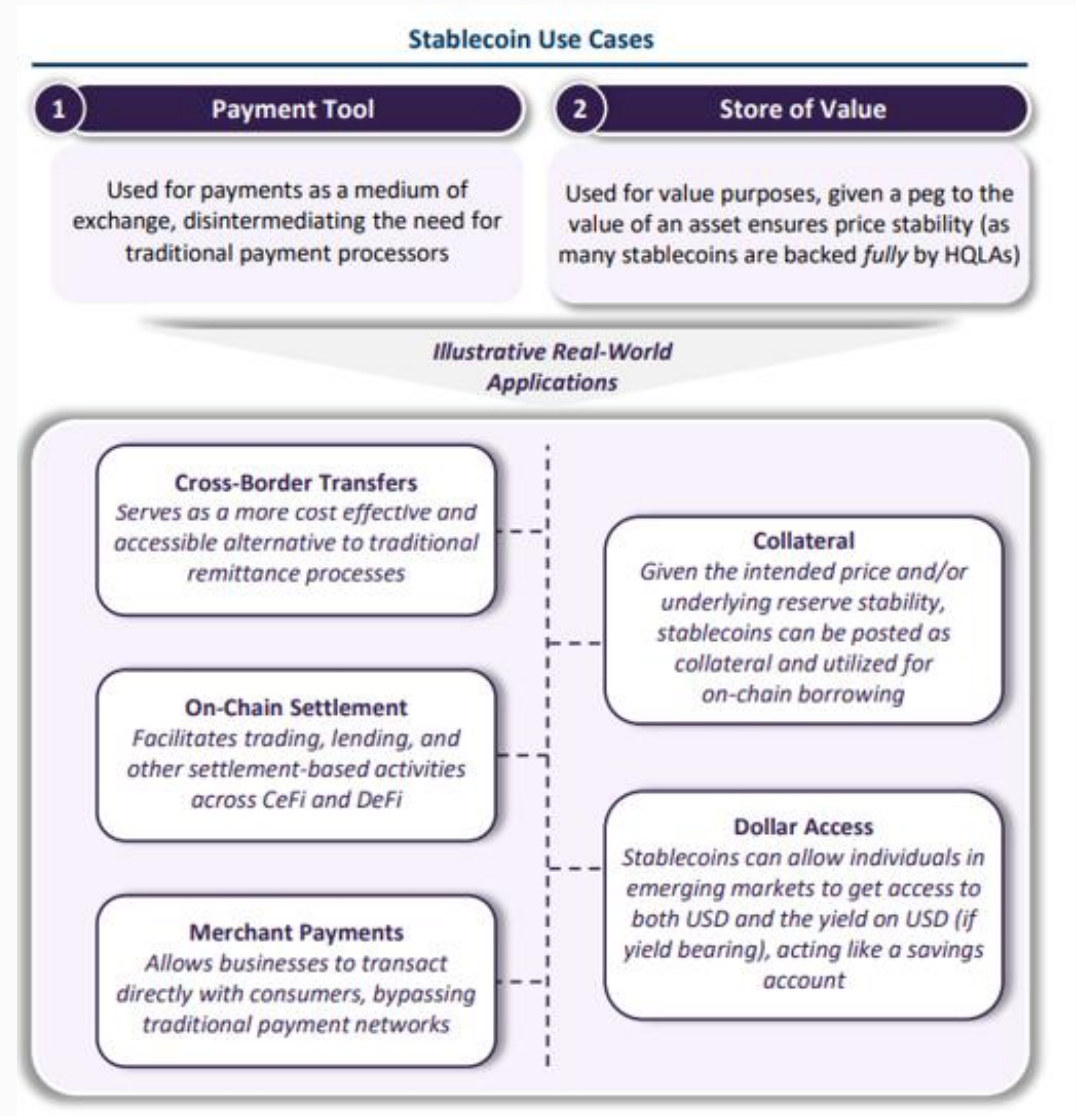
3 “Currency” : Use Cases

Mentioned by department of treasury

- Cross-Boarder Transfers
- Collateral
- On-Chain Settlement
- Merchant Payment
- Dollar Access

What' s the potential?

- Tax/Tariff auto-collection
- Retail sell of Treasury Bills
- RWA (Real-world assets) to be traded on chain, allowing illiquid assets to be publicly traded.



3 “Currency” : US Legislative Stablecoin Framework

Current US Legislative Stablecoin Framework Proposal²

On March 13, 2025, the US Senate Banking Committee voted in favor of the GENIUS Act. The bill, introduced by **Senator Bill Hagerty (R)**, establishes a regulatory framework for US issued stablecoins and creates a definition for the term “payment stablecoin”

Definition	■ Definition: A digital asset designed to be used as a means of payment, which the issuer is obligated to redeem for fixed amount, and which represents (or creates reasonable expectation) that it will maintain a stable value relative to a fixed amount ■ Exclusions: cannot offer yield or interest, not a national currency, security, commodity, or a deposit (including a blockchain-based deposit)
Issuer	■ Regulatory Authority: Federal or State (but Federal regulators determine licensing process for all issuers). Can be approved as a: — Subsidiary of a bank, or — Non-bank issuer (Fed or State regulated if less than \$10bn M/C)
Reserves	■ 1:1 backing: USD coins and currency (including Federal reserve notes); funds held in demand deposits, treasury bills, notes or bonds with a remaining / issued with maturity of 93 days or less; repurchase agreements with maturity of 7 days or less backed by T-Bills with a maturity of 90 days or less; reverse repurchase agreements with maturity of 7 days or less collateralized by Treasury notes, bills or bonds; money market funds that invest in the aforementioned; or central bank reserve deposits ■ Monthly disclosure requirements
Blockchain	■ Public blockchains only
Other	■ Legal control: Issuers must be able to freeze, burn or prevent transfer of stablecoins if required by law ■ All permitted issuers are subject to Fed laws applicable to financial intuitions for the purposes of Bank Secrecy Act and are required to establish BSA/AML compliant programs

Assignment 1 (Regular Track): Crypto Market Analysis

Lecture 1

1. (10) Download data from <https://coinmarketcap.com/historical/> by hand OR through APIs from any crypto exchange you prefer (e.g. coinbase, binance, etc.).
2. (10) Check and download data from <https://www.blockchain.com/explorer/charts> , either by hand or through API is fine.
3. (30) List the data you think that can be used to predict BTC prices and explain the reasons behind each. Quantitative explanations are preferred.

Lecture 2

4. (10) Find and implement a limit order book simulator on Github. Display and explain its outcomes.

Lecture 3

5. (10) Locate the crypto asset(s) you want to trade. Describe its statistical features, risk factors, and other information you think is important.
6. (30) What kind of strategies do you think could be applied to the asset(s) you selected? Explain how it is related to your strategy.

Assignment 1 (Express Track): Crypto Market Analysis

Challenge 1(100): Implement the factor model from the paper: Liu, Y., Tsyvinski, A., & Wu, X. (2022). Common risk factors in cryptocurrency. The Journal of Finance, 77(2), 1133–1177.

The paper's factor model excludes blockchain-specific variables because it is designed as a general asset-pricing framework rather than a BTC-focused predictor. Can you combine both blockchain factors and price factors together for BTC and ETH? This problem serves as preparation for designing market-neutral strategies.

Challenge 2(100): Write a simulator for automated market maker (AMM) from Uniswap. Compare the price/liquidity generated from the traditional simulator against the AMM. The purpose of this problem is to prepare for developing a cross-exchange arbitrage strategy.

Appendix:

Play with blockchain demo

<https://andersbrownworth.com/blockchain/blockchain>

Real Blockchain

<https://www.blockchain.com/explorer/blocks/btc/mempool.space>

References:

Blockchain

- Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system.
- Yaga, D., Mell, P., Roby, N., & Scarfone, K. (2019). Blockchain technology overview. arXiv preprint arXiv:1906.11078.
- <https://www.pwc.com/us/en/industries/financial-services/library/instant-payments-blockchain-for-banks.html>
- <https://adilmoujahid.com/posts/2018/03/intro-blockchain-bitcoin-python/>

Stablecoin

- <https://home.treasury.gov/system/files/221/TBACCharge2Q22025.pdf>

A minimalist geometric design featuring a large, light gray rounded rectangle with a thin dark gray border. The rectangle has rounded corners on the top-left and bottom-right. Two dark gray circles are positioned on the left and right sides of the rectangle, each connected to the border by a thin dark gray line. The text "Thank You" is centered within the rectangle in a dark gray, monospace-style font. Below the text is a solid dark gray horizontal bar.

T h a n k Y o u

